

Computer Graphics – April 8th, 2021

Name: _____

Seat Number: _____

Matriculation Number: _____

Please read the following instructions carefully:

- You have sixty (60) minutes to complete the exam.
- You can answer in German or in English.
- Check that the exam is complete (22 pages on 11 sheets) and readable!
- Please use a blue or black ballpen, not a red or green one. Do not use a pencil!
- Only text on the stapled sheets will be evaluated.
- If needed, we staple additional blank sheets.
- Only writing material is allowed!

I have read and understood the instructions above.

Erlangen, April 8th, 2021

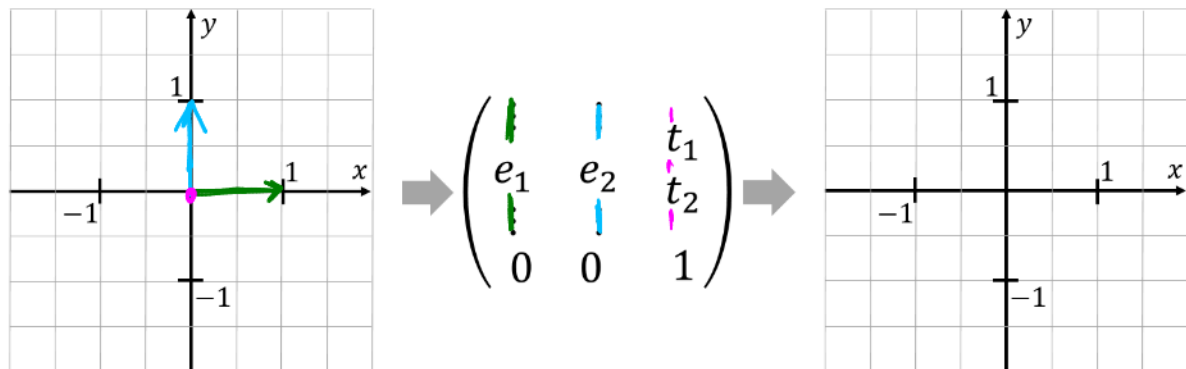
Signature

1	2	3	4	5	6	ALL
12	14	10	12	6	6	60

Grade:

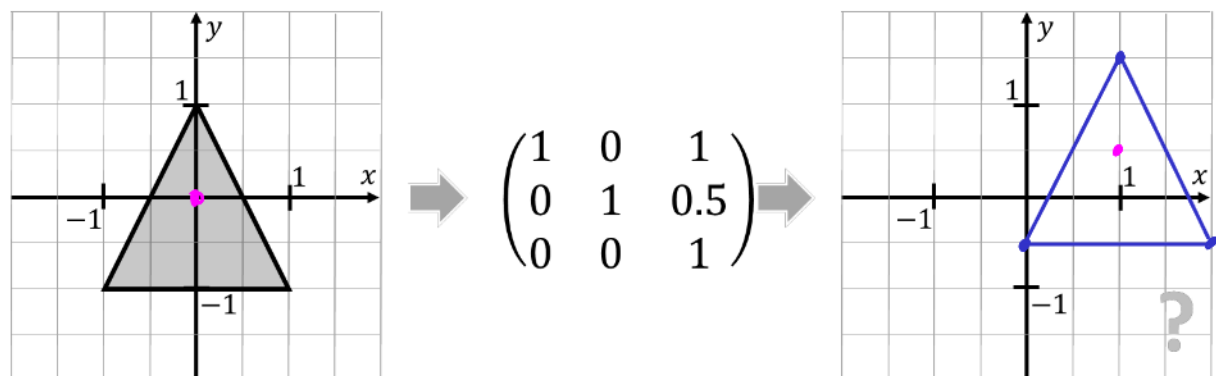
Assignment 1: Affine Mappings in 2D [12 points]

Affine mappings in 2D can be expressed using homogeneous coordinates and 3×3 matrices with the following structure, where (e_1, e_2) is the **linear part** (image of base vectors) and (t_x, t_y) is the **translation**:

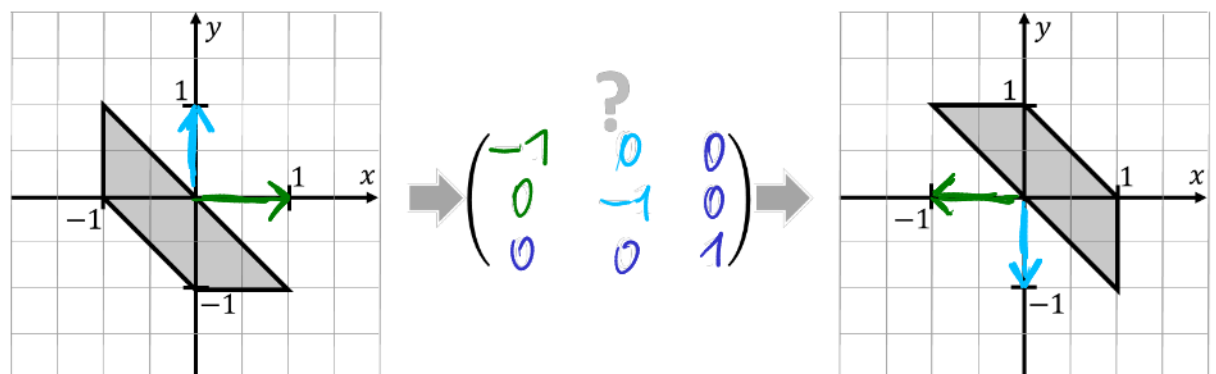


Each of the sub-assignments a) to e) deals with one mapping (in the coordinate system depicted above) and consists of two images (before and after mapping) and one matrix that has the desired effect. Complete each of the sub-assignments by filling in the missing part marked with a question mark.

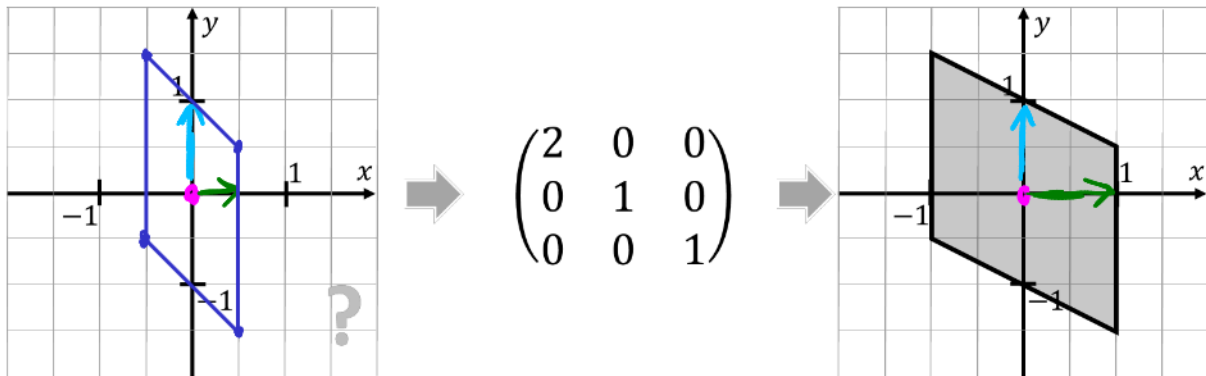
a) [1.5 P]



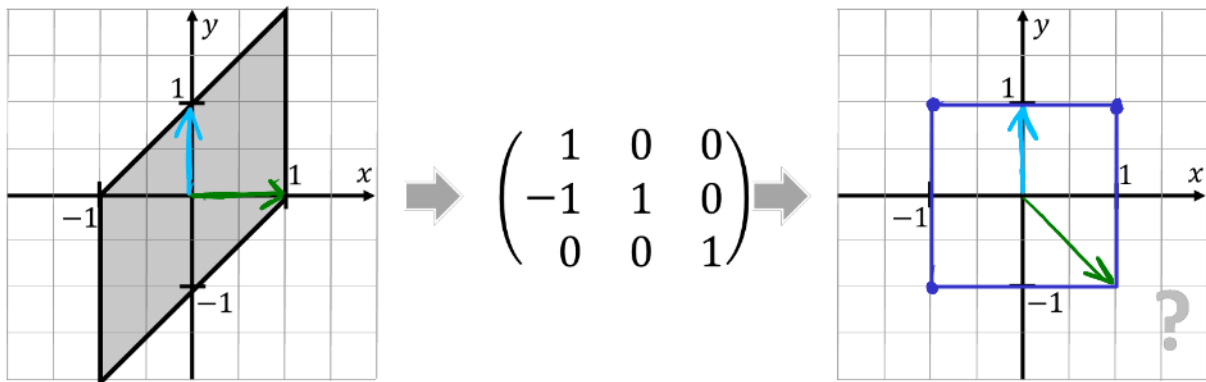
b) [1.5 P]



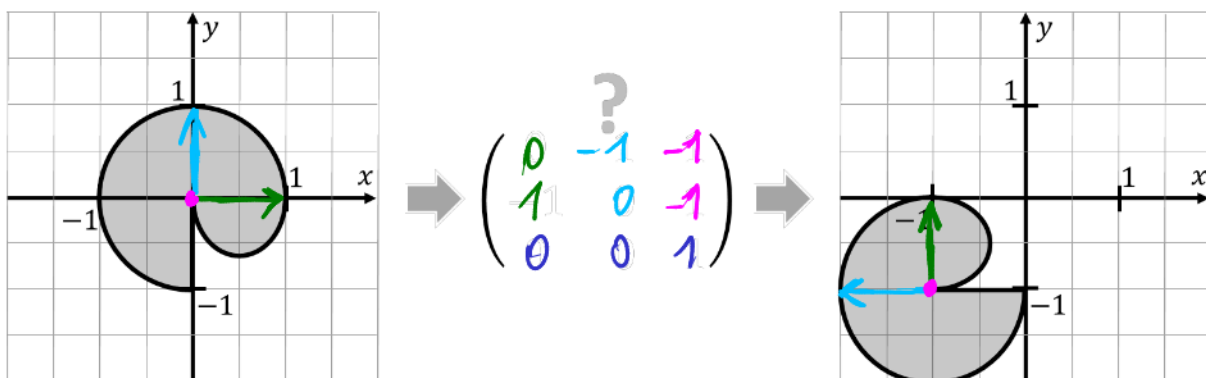
c) [2 P]



d) [2 P]

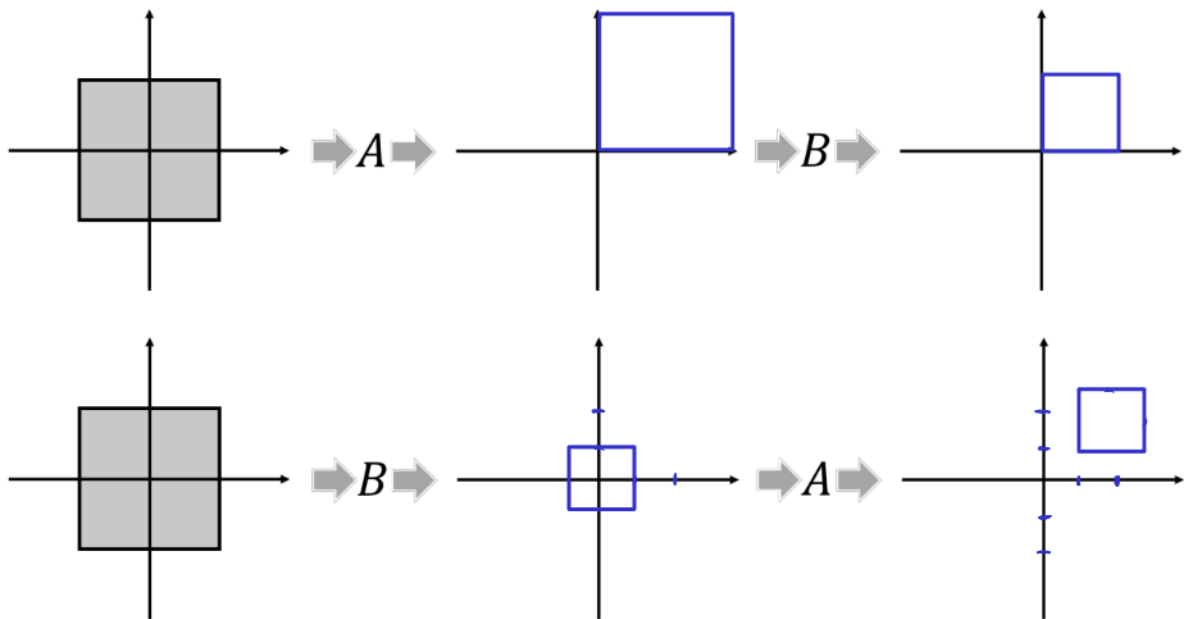


e) [2 P]



$$A = \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

- f) [3 P] Think of one mapping A which is **rigid, but not linear**, and another mapping B which is **linear, but not rigid**. In the diagrams below, apply your two mappings to the unit square one after another (first row: first A , then B ; second row: first B , then A)!



In the table below, each row describes one of the mappings (either a single mapping or a combination). The properties demanded above are already stated (grey grid cells). Fill in the empty (white) grid cells by using a tick mark (✓) if the respective mapping is rigid/linear/affine or a cross mark (✗) if it is not.

	rigid	linear	affine
A	✓	✗	✓
B	✗	✓	✓
AB	✗	✗	✓
BA	✗	✗	✓

Assignment 2: Perspective Projection & Texture Mapping [14 points]

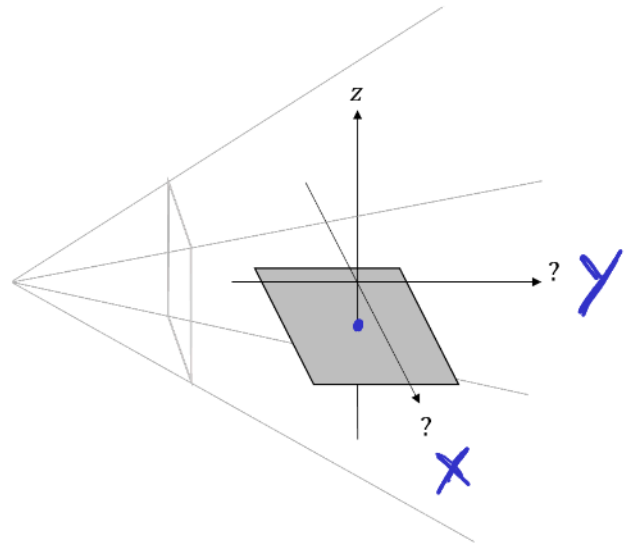
Given is the square $ABCD$ with the four vertices

$$A = (-1, -1, -1), B = (-1, 1, -1),$$

$$C = (1, -1, -1), D = (1, 1, -1).$$

We project the square perspectively using the following homogeneous matrix:

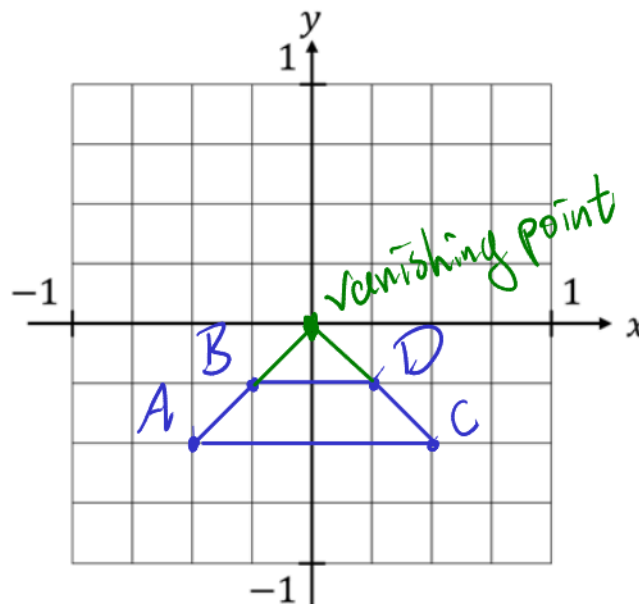
projection $M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 3 \end{pmatrix}$



- a) **[4 P]** Map the vertices to image space using matrix M and fill in the table with the mapping results. For all vertices, give both their two-dimensional image coordinates and their depth value, both after dehomogenization!

Vertex	Image coordinates $\in [-1, 1]^2$	Depth $\in [-1, 1]$
A	$-\frac{1}{2}, -\frac{1}{2}$	$-\frac{1}{2}$
B	$-\frac{1}{4}, -\frac{1}{4}$	$\frac{1}{4}$
C	$\frac{1}{2}, -\frac{1}{2}$	$-\frac{1}{2}$
D	$\frac{1}{4}, -\frac{1}{4}$	$\frac{1}{4}$

- b) **[2 P]** Sketch the resulting shape in image space into the following 8x8 pixel grid:



② a)

$$M\tilde{A} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ -1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ -1 \\ 2 \end{pmatrix}$$

perspective divide

$$\begin{pmatrix} -1/w \\ -1/w \\ -1/w \end{pmatrix} = \begin{pmatrix} -1/2 \\ -1/2 \\ -1/2 \end{pmatrix} \begin{matrix} x \\ y \\ \text{depth} \end{matrix}$$

homogenized A
with 1 as 4th
component

$$M\tilde{B} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 1 \\ 4 \end{pmatrix} \longrightarrow \begin{pmatrix} -1/4 \\ -1/4 \\ 1/4 \end{pmatrix}$$

$$M\tilde{C} = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1/2 \\ -1/2 \\ -1/2 \end{pmatrix}$$

$$M\tilde{D} = \begin{pmatrix} 1 \\ -1 \\ 1 \\ 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 1/4 \\ -1/4 \\ 1/4 \end{pmatrix}$$

c) [2 P] What is the view direction?

$$V = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

d) [2 P] What is the **vanishing point** of lines parallel to the viewing direction? Give the answer in two ways. First, construct the vanishing point **graphically** using the grid in b). Second, **compute** this vanishing point from the matrix!

$$f = \lim_{x \rightarrow \infty} v = \begin{pmatrix} 0 \\ x \\ 0 \end{pmatrix} \quad \text{"far away in view direction"}$$

$$Mf = \lim_{x \rightarrow \infty} \begin{pmatrix} 0 \\ 0 \\ x \\ x+3 \end{pmatrix} \xrightarrow{\text{perspective divide}} \lim_{x \rightarrow \infty} \begin{pmatrix} 0 \\ 0 \\ \frac{x}{x+3} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \} \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ "vanishing point"}$$

Better homogeneous component set to 0 $\Rightarrow$ point at infinity

$$M \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} \xrightarrow{\text{perspective divide}} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \} \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ "vanishing point"}$$

② c) From the sketch of the resulting shape one can see:

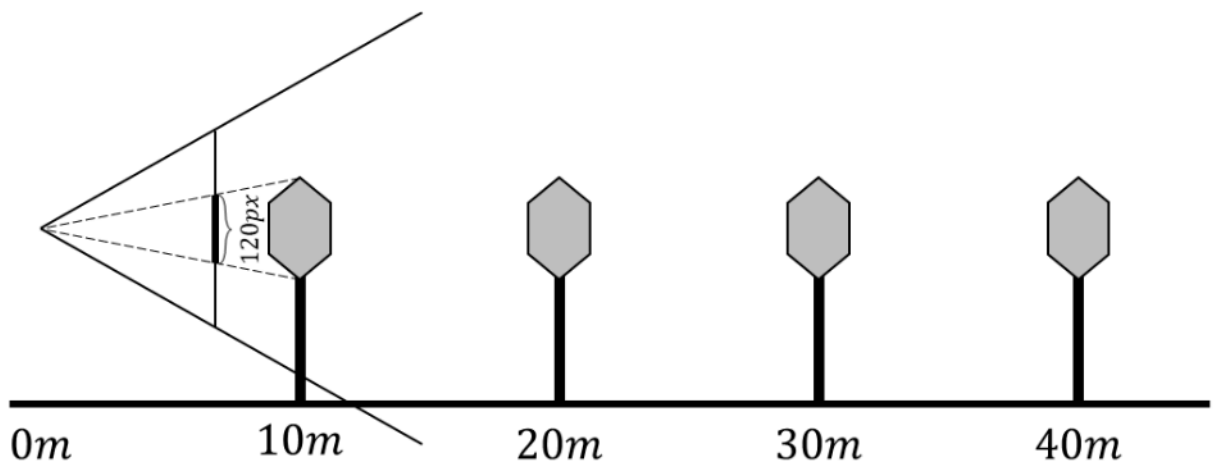
$$V = \frac{B-A}{|B-A|} = \frac{D-C}{|D-C|} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Alternative: Bottom row of $M \rightarrow$ perspective divide is done in view direction

$$\begin{pmatrix} \vdots \\ 0 & 1 & 0 & 3 \end{pmatrix}$$

$x \quad y \quad z$

- e) **[4 P]** Given is the following scene with four equivalent traffic signs within the depicted view frustum. In the perspective image, the leftmost traffic sign (in 10m distance to the camera) is 120 pixels tall (only the sign, without the post).



The other traffic signs in the image are of the same height in world space and located at 20m, 30m and 40m, respectively. How tall are those **in image space**?

10m: 120 px 20m: 60 px 30m: 40 px 40m: 30 px	perspective divide $\frac{1}{\text{depth}}$ $\Rightarrow$ double distance, half the size
--	--

A texture is mapped to the traffic signs. For the first sign **at 10m**, the level of detail algorithm decides that MIP map **level 8** is optimal. For each of the other three traffic signs, state which level(s) will be used (smaller level means coarser texture)!

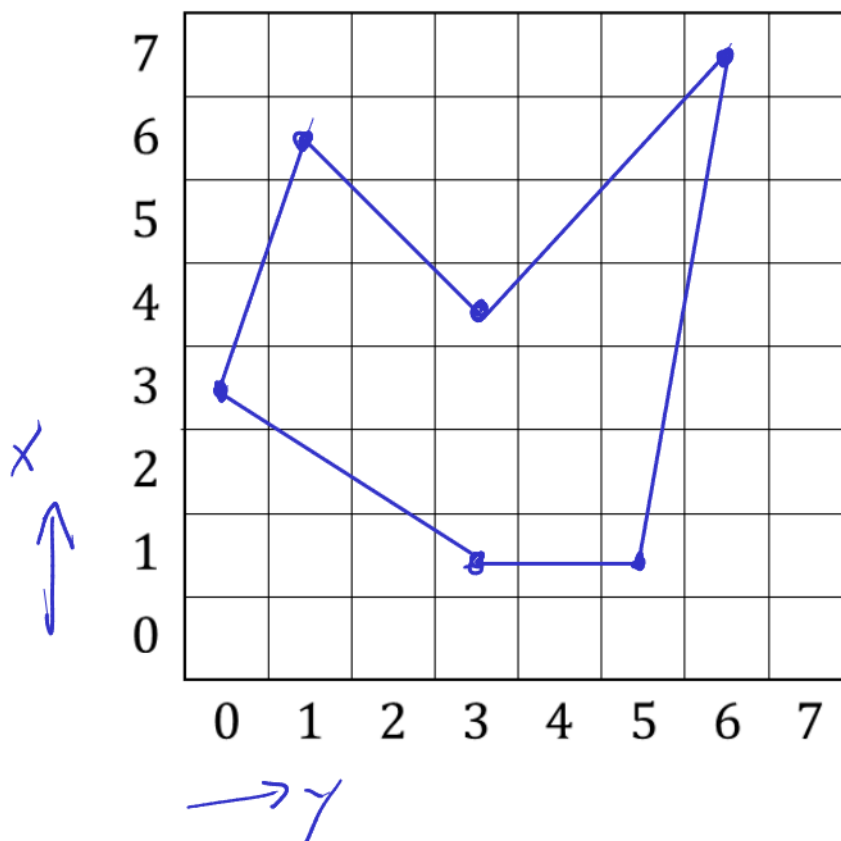
10m: 8 20m: 7 30m: "between 6 and 7" = $8 + \log_2(\frac{1}{3}) = 8 - \log_2(3)$ 40m: 6
--

Assignment 3: Polygon Rasterization [10 points]

- a) **[6 P]** The following image shows an **Edge Table for a polygon**. Sketch the corresponding polygon into the grid below the Edge Table! You do not have to fill single pixels, just sketch the outline of the polygon!

↪ Scanline algorithm

	y_{lower}	x_{lower}	y_{upper}	$1/m$	$next$
✓	0	3	1	3	
✓	0	3	3	$-2/3$	
✓	1	6	3	-1	
✓	3	1	5	0	
✓	3	4	6	1	
✓	5	1	6	6	



- b) [2 P] How could you incorporate wireframe rendering (drawing only the outlines of polygons) into the scanline algorithm?

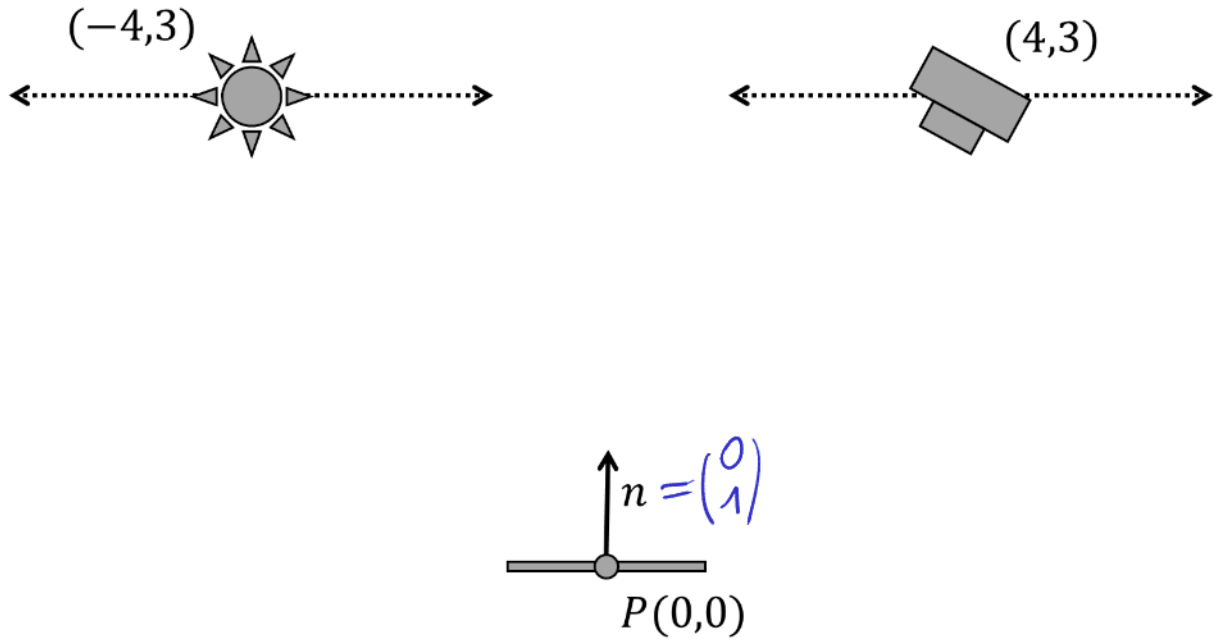
- Draw only first and last pixel of lines that are drawn in each scanline
- If $\frac{1}{m} > 1$ there may be wholes in the wireframe
 To solve this: If $\frac{1}{m} > 1$, then fill the $(m-1)$ pixel left of the line


- c) [2 P] An alternative to polygon rasterization using the scanline algorithm is polygon rasterization via testing a pixel against the edge equations of all lines forming the polygon. Is this a viable option for the polygon from a)? Why/why not? What would have to be done to make it infeasible/feasible to use edge equation tests for this polygon?

No, because the polygon is not convex!
One can split it up into two convex polygons and rasterize those.

Assignment 4: Phong Lighting & Shading [12 points]

The image below shows a point P at $(0,0)$ belonging to a surface with the normal vector $(0,1)$. The surface is lit by a point light located at $(-4,3)$ and captured by a camera located at $(4, 3)$. The light coming from the point light is not attenuated. For subtasks a) and b), you can ignore the dashed arrows.



- a) **[3 P]** For the point P , write down the respective light vector l , view vector v and reflection vector r !

$$l = \begin{pmatrix} -4 \\ 3 \end{pmatrix} \frac{1}{\sqrt{25}} = \begin{pmatrix} -4/5 \\ 3/5 \end{pmatrix} = \begin{pmatrix} -0,8 \\ 0,6 \end{pmatrix}$$

$$v = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \frac{1}{\sqrt{25}} = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix} = \begin{pmatrix} 0,8 \\ 0,6 \end{pmatrix}$$

$$r = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \frac{1}{\sqrt{25}} = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix} = \begin{pmatrix} 0,8 \\ 0,6 \end{pmatrix}$$

- b) [5 P] Let's assume the following Phong Lighting parameters for the surface:

$$k_{amb} = (0,0,0), k_{diff} = (0,0,1), k_{spec} = (1,0,0), n_{shiny} = 2$$

The point light is of intensity $I_{in} = (1,1,1)$.

Compute the color of point P!

$$\begin{aligned} L &= L_{amb} + L_{diff} + L_{spec}, \quad \cos = 0,6, \quad ror = 1 \\ &= k_{amb} I_{amb} + k_{diff} I_{in} (\cos) + k_{spec} I_{in} (ror)^{n_{shiny}} \\ &= 0 + \begin{pmatrix} 0 \\ 0 \\ 0,6 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 0 \\ 0,6 \end{pmatrix} \end{aligned}$$

- c) [4 P] Next, assume that both the point light and the camera can move along the dashed lines.

When **one** of the two components **moves**, the **other** stays at its **starting point**.

For both the diffuse and the specular part, state what happens for the four movements in the tables below.

First, how does the diffuse part change for each of the following four position changes? In each row, mark one of the possibilities!

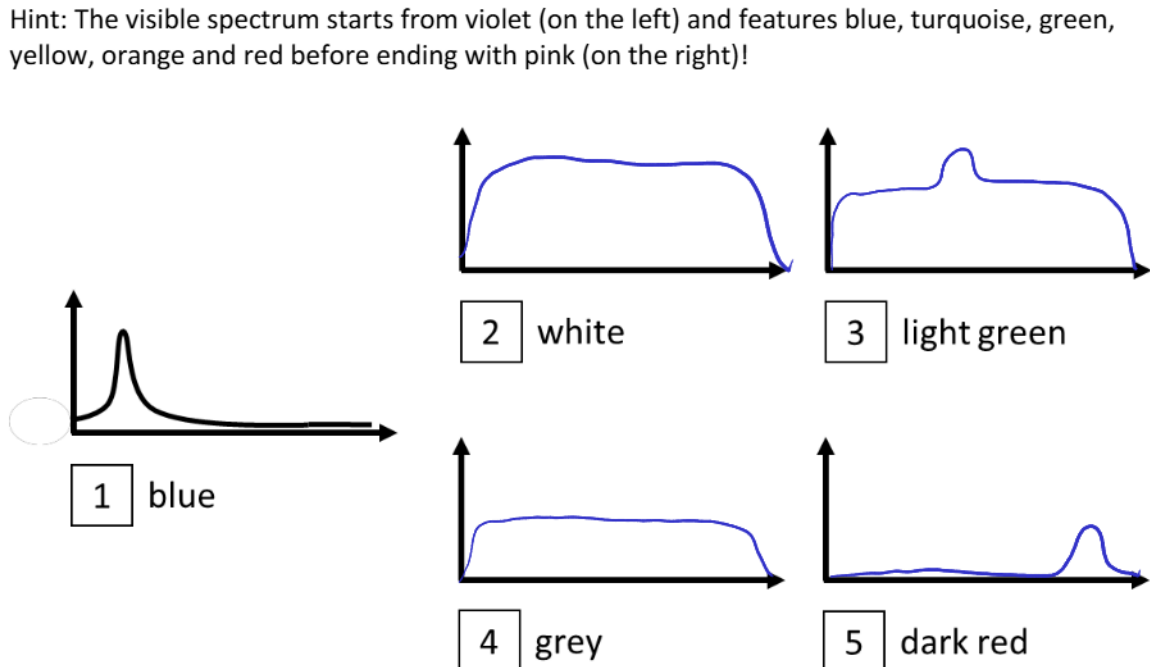
		diffuse does not change	diffuse gets darker	diffuse gets brighter
light	moves left		X	
	moves right			X
camera	moves left	X		
	moves right	X		

Second, how does the specular part change for each of the following four position changes? In each row, mark one of the possibilities!

		specular does not change	specular gets darker	specular gets brighter
light	moves left		X	
	moves right		X	
camera	moves left		X	
	moves right		X	

Assignment 5: Color Models [6 points]

- a) **[2 P]** Colors can be characterized by their spectrum, a function of wavelengths the visible color is made up of. For the first color below, the spectrum is already given. For each of the other colors, sketch the function of wavelengths which leads to this color impression!



- b) **[4 P]** Below you can find representations of the five colors from a) in each of the three color models RGB, CMYK and HSV. Fill in the respective numbers to link all representations to the correct color. For blue, the solution is already given.

Hint: In HSV, the angles for red, green and blue are 0° , 120° and 240° , respectively.

RGB	CMYK	HSV
1 (0, 0, 1)	2 (0, 0, 0, 0)	4 (0°, 0, 0.5)
5 (0.5, 0, 0)	1 (1, 1, 0, 0)	1 (240°, 1, 1)
2 (1, 1, 1)	4 (0, 0, 0, 0.5)	3 (120°, 0.5, 1)
3 (0.5, 1, 0.5)	5 (0, 0.5, 0.5, 0.5)	2 (0°, 0, 1)
4 (0.5, 0.5, 0.5)	3 (0.5, 0, 0.5, 0)	5 (0°, 1, 0.5)

Assignment 6: Raytracing [6 points]

Given is a scene with k spheres. For rasterization, each sphere is subdivided into n triangles. We want to render a view of the scene with p pixels. The spheres cover most of the image with only little occlusion.

Provide the time complexity of the rendering process for the following setups:

a) **[3 P]** Rasterization using the triangles described above:

- vertex shader:

$$O(kn)$$

VS $\rightarrow$ rasterizer $\rightarrow$ PS

- pixel shader:

$$O(p)$$

b) **[1.5 P]** Ray Casting with a direct ray-sphere intersection test in $O(1)$:

$$O(pk)$$

c) **[1.5 P]** Ray Casting on a well-chosen bounding volume hierarchy of the triangles described above:

$$O(p \cdot \log k)$$

building the hierarchy $O(k \log k)$